



LOCAL AI

Local LLM Deployment & Tuning

Deploying high-throughput open-weight models locally with Ollama, vLLM, and QLoRA tuning.

EXECUTIVE SUMMARY

Running private AI models on local workstations eliminates per-token API fees while protecting company data. This guide covers GPU VRAM sizing, quantization trade-offs, continuous batch serving, and QLoRA fine-tuning.

1 High-Throughput Local Serving Architecture

Production deployment uses vLLM or Ollama with PagedAttention to maximize GPU utilization. By serving models in 4-bit (AWQ / Q4_K_M) quantization, large 32B–70B models run comfortably on consumer workstations.



2 Production Implementation & Schema

```
# Serving Models with vLLM (Linux / WSL2)
python3 -m vllm.entrypoints.openai.api_server \
  --model Qwen/Qwen2.5-Coder-32B-Instruct-AWQ \
  --quantization awq \
  --gpu-memory-utilization 0.92 \
  --max-model-len 16384
```



3 Parameter-Efficient Fine-Tuning (QLoRA)

Rather than modifying all model parameters, QLoRA freezes the 4-bit quantized base model and trains lightweight low-rank adapter layers, reducing required training VRAM by over 75%.

vllm-server • Performance Monitor

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```
[INIT] Model loaded: Qwen2.5-Coder-32B-Instruct-AWQ
✓ [VRAM] 19.4 GB allocated across 24 GB GPU memory
✓ [BENCH] 64.2 tokens/second sustained throughput
✓ [COMPAT] OpenAI-compatible API running at http://localhost:8000/v1
[PRIVACY] 100% offline private inference • Zero telemetry
```

4 Key Takeaways & Best Practices

- ✓ **VRAM Budgeting**

Target 4-bit quantization (AWQ or Q4_K_M) to run 32B parameter models smoothly on 24GB VRAM graphics cards.
- ✓ **Continuous Batching**

Use vLLM serving to achieve 4x–8x higher concurrent request throughput compared to naive inference.
- ✓ **LoRA Adapter Swapping**

Train separate lightweight LoRA adapters for different domains without duplicating base model weights.
- ✓ **Open API Standard**

Expose local models using standard OpenAI-compatible endpoints so existing libraries connect seamlessly.

IMPLEMENTATION SUPPORT

Need engineering assistance integrating these architectures into your business? Contact our team directly at futureai4all@gmail.com.